

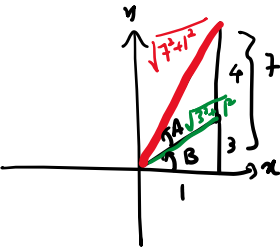
Q	Solution	Comments
1a	$(y-5)(y^2+5y+25)$	- Recognise cubic expressions of form $a^3 \pm b^3$ and factorise using $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$ - Alternative method to apply Factor Theorem to obtain linear and quadratic factor is acceptable but less efficient
1b	$7875 + 125 = 8000$ $y^3 = 8000$ $y = 20$ $\therefore 7875 = (20-5)(400+100+25)$ $= (15)(525)$	For 'Hence' questions, the connection to the previous part must be clearly shown in their working
2	$\frac{1}{4}(28+\sqrt{2}) = \frac{1}{2}(5+\sqrt{2})(YZ)\left(\frac{\sqrt{2}}{2}\right)$ $YZ = \frac{28+\sqrt{2}}{5\sqrt{2}+2}$ $= \frac{(28+\sqrt{2})(5\sqrt{2}-2)}{25(2)-4}$ $= \frac{140\sqrt{2}-56+10-2\sqrt{2}}{46}$ $= (3\sqrt{2}-1) \text{ cm}$	- Draw diagram to visualise problem to ensure formula is used correctly TIP: Simplify fractions first before rationalising denominator, else it may result in overcomplicated working and room for careless mistakes, especially if $\sin 45^\circ = \frac{1}{\sqrt{2}}$ is used. Common Errors - Missing brackets e.g. $\frac{28+\sqrt{2}(5\sqrt{2}-2)}{25(2)-4}$ when it's supposed to be $\frac{(28+\sqrt{2})(5\sqrt{2}-2)}{25(2)-4}$
3	$\begin{array}{r} 3x-1 \\ 2x^2-9x-5 \overline{) 6x^3-29x^2+4x-23} \\ \underline{-(6x^3-27x^2-15x)} \\ -2x^2+19x-23 \\ \underline{-(-2x^2+9x+5)} \\ 10x-28 \end{array}$ $\frac{6x^3-29x^2+4x-23}{2x^2-9x-5} = 3x-1 + \frac{10x-28}{(2x+1)(x-5)}$ $\frac{10x-28}{(2x+1)(x-5)} = \frac{A}{2x+1} + \frac{B}{x-5}$ $10x-28 = A(x-5) + B(2x+1)$ When $x = 5$, $22 = 11B$ $B = 2$ When $x = -\frac{1}{2}$, $-5-28 = -\frac{11}{2}A$ $A = 6$ $\frac{6x^3-29x^2+4x-23}{2x^2-9x-5} = 3x-1 + \frac{6}{2x+1} + \frac{2}{x-5}$	- Recognise improper algebraic fractions by looking at degree of numerator vs degree of denominator. - Long division common errors: - Stopping when remainder is quadratic (divide until degree of remainder is strictly less than remainder of divisor) - Mistakes in subtracting at each step (put a bracket around algebraic expression to be subtracted to avoid this mistake) - After long division, express improper fraction as quotient + $\frac{\text{remainder}}{\text{divisor}}$

4a	$ax^2 + 7x + b = a \left(x^2 + \frac{7}{a}x + \frac{b}{a} \right)$ $= a \left[\left(x + \frac{7}{2a} \right)^2 - \frac{49}{4a^2} + \frac{b}{a} \right]$ $= a \left(x + \frac{7}{2a} \right)^2 - \frac{49}{4a} + b$ $\therefore -\frac{49}{4a} + b \leq 8$ $b \leq 8 + \frac{49}{4a}$ & $a < 0$ Alternative solution 1 $y = ax^2 + 7x + b - 8$ $b^2 - 4ac = 49 - 4a(b - 8)$ $49 - 4a(b - 8) \leq 0$ $\frac{49}{4a} \geq b - 8 \quad (\because a < 0)$ $b \leq 8 + \frac{49}{4a}$ & $a < 0$ Alternative solution 2 $y = ax^2 + 7x + b$ $\frac{dy}{dx} = 2ax + 7$ When $\frac{dy}{dx} = 0$, $x = -\frac{7}{2a}$ $\therefore a \left(-\frac{7}{2a} \right)^2 + 7 \left(-\frac{7}{2a} \right) + b \leq 8$ $b \leq 8 + \frac{49}{4a}$ & $a < 0$	Common errors: When conditions on unknown constants are required, inequalities involving the unknown constants are required where the unknown constants is the subject (e.g. $-\frac{49}{4a} + b \leq 8$ is not sufficient, $b \leq 8 + \frac{49}{4a}$ is needed)
4b	Any negative a And corresponding value of b where $b \leq 8 + \frac{49}{4a}$ $a = -2$, e.g. $b \leq 1.875$ $\therefore b = 1$	EXAM TIP: Gain partial credit by at least giving a possible value of a, which is easy as any negative a will be a possible value.

5a	When $x = \frac{\pi}{3}, y = 0$ $\frac{dy}{dx} = \frac{(3\cos 3x)(\cos x - 2) - (-\sin x)\sin 3x}{(\cos x - 2)^2}$ $= \frac{(3\cos 3x)(\cos x - 2) + (\sin x)(\sin 3x)}{(\cos x - 2)^2}$ $\left. \frac{dy}{dx} \right _{x=\frac{\pi}{3}} = \frac{3(-1)\left(-\frac{3}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)(0)}{\frac{9}{4}}$ $= 2$ $\therefore$ equation of tangent is $y - 0 = 2\left(x - \frac{\pi}{3}\right)$ $y = 2x - \frac{2\pi}{3}$	Common errors: - Careless or remember quotient/product/chain rule/differentiation of trigonometric functions wrongly - Wrong value of trigo ratios of special angles - Not leaving working in exact form, resulting in non-exact intermediate working and hence final answer. - Presentation: When answer is to be given in equation form, gradient / y-intercept to be expressed in exact form. However, if question asks for coordinates, then answer to be expressed in 3sf unless otherwise stated.
5b	$Q\left(0, -\frac{2\pi}{3}\right)$	
6	$V = \frac{4}{3}\pi r^3$ $\frac{dV}{dr} = 4\pi r^2$ $\left. \frac{dV}{dr} \right _{r=15} = 900\pi$ $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$ $42 = 900\pi \times \frac{dr}{dt}$ $\frac{dr}{dt} = \frac{7}{150\pi}$ $A = 4\pi r^2$ $\frac{dA}{dr} = 8\pi r$ $\left. \frac{dA}{dr} \right _{r=15} = 120\pi$ $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ $= 120\pi \times \frac{7}{150\pi}$ $= 5.6 \text{ cm}^2/\text{s}$	Common errors: $\frac{42}{900\pi} = \frac{42}{900}\pi$ $\frac{dV}{dr} = \frac{4}{3}\frac{\pi r^3}{r}$ (misconception that $\frac{dV}{dr}$ is fraction)

7a	$\log_3 p - \log_{27} p = -4$ $\log_3 p - \frac{\log_3 p}{\log_3 3^3} = -4$ $\log_3 p - \log_3 p^{\frac{1}{3}} = -4$ $\log_3 p^{\frac{2}{3}} = -4$ $\log_3 p = -6$ $p = 3^{-6}$ $= \frac{1}{729}$	For logarithmic expressions involving algebra, simplify using power law first to avoid equating to complicated algebraic expressions later [e.g. $\log_3 p - \frac{\log_3 p}{\log_3 3^3} = -4$ $\log_3 p - \frac{\log_3 p}{3} = -4$ $3\log_3 p - \log_3 p = -12$ $\log_3 p^3 - \log_3 p = -12$ $\log_3 p^2 = -12$ $p^2 = 3^{-12}$ $p = \pm\sqrt{3^{-12}} \text{ (reject } -\sqrt{3^{-12}} \text{ since } \log_3 p \text{ will be undefined)}$ $p = 3^{-6}$ $= \frac{1}{729}$ The decision not to simplify $\log_3 p^2$ to $2\log_3 p$ resulted the need to reject negative solutions]
7bi	$\ln qr = \ln(q + r + 1)$ $qr = q + r + 1$ $qr - q = r + 1$ $q = \frac{r+1}{r-1}$	
7bii	$q > 0 \text{ for } \ln q \text{ to be defined}$ $r > 0 \text{ for } \ln r \text{ to be defined}$ $\therefore r + 1 > 0$ $q = \frac{r+1}{r-1}, \text{ since } q > 0 \text{ and } r + 1 > 0$ $\therefore r - 1 > 0$ $\Rightarrow r > 1$	- Concluding $q > 0$ means denominator is > 0 is insufficient. $q > 0$ alone can only mean either - both numerator and denominator > 0 - both numerator and denominator < 0 In order to explain $r > 1$, we need to explain why only case 1. is true and not case 2. Hence, there is a need to state explicitly first why numerator > 0, which comes from the fact that $r > 0$

8a	$\begin{aligned} \text{LHS} &= \frac{\sin x}{\cot x + \sin x - \operatorname{cosec} x} \\ &= \frac{\sin x}{\left(\frac{\cos x}{\sin x}\right) + \sin x - \left(\frac{1}{\sin x}\right)} \\ &= \frac{\sin x}{\left(\frac{\cos x + \sin^2 x - 1}{\sin x}\right)} \\ &= \frac{\sin^2 x}{\cos x + \sin^2 x - 1} \\ &= \frac{1 - \cos^2 x}{\cos x - \cos^2 x} \\ &= \frac{(1 + \cos x)(1 - \cos x)}{\cos x(1 - \cos x)} \\ &= \frac{1 + \cos x}{\cos x} \\ &= \sec x + 1 \\ &= \text{RHS} \end{aligned}$	Common errors: • $\frac{1}{A+B+C} = \frac{1}{A} + \frac{1}{B} + \frac{1}{C}$ • $\frac{1}{\frac{A}{B} + \frac{C}{D}} = \frac{B}{A} + \frac{D}{C}$ • Missing to factorise $1 - \cos^2 x$
8b	$\begin{aligned} \frac{\sin 2x}{\cot 2x + \sin 2x - \operatorname{cosec} 2x} &= \tan^2 2x \\ 1 + \sec 2x &= \sec^2 2x - 1 \\ \sec^2 2x - \sec 2x - 2 &= 0 \\ (\sec 2x + 1)(\sec 2x - 2) &= 0 \\ \sec 2x &= -1 \text{ or } \sec 2x = 2 \\ \cos 2x &= -1 \text{ or } \cos 2x = 0.5 \\ \text{Since } 0^\circ < x < 90^\circ, 0^\circ < 2x < 180^\circ. \\ 2x &= 180^\circ (\text{rej.}) \text{ or } \text{Basic angle} = \cos^{-1}(0.5) = 60^\circ \\ 2x &= 60^\circ \text{ or } 300^\circ (\text{rej.}) \\ \therefore x &= 30^\circ \end{aligned}$	Common errors: • Copying from (a) but did not change angle to $2x$ • Range for $2x$ not stated • Not finding all angles in relevant quadrants, conveniently missing them out just because they're outside the range. There is a need to state the angles in the 4 quadrants that will solve the trigonometric equation and then rejecting based on the range for $2x$ • Solving $\cos 2x = -1$ by finding basic angle, which is incorrect. To solve trigonometric equations equal to 0, 1 or -1, we have to use the trigonometric graph
9a	sub. $x = 1$ & $y = -1$ into $y = x^4 - 4x^3 + ax^2 + bx + 8$, $a = -b - 6 \text{ -----(1)}$ $\frac{dy}{dx} = 4x^3 - 12x^2 + 2ax + b$ At $x = 1, \frac{dy}{dx} = 0$ $\therefore b = 8 - 2a \text{ -----(2)}$ sub. (1) into (2), $b = -20$ $\therefore a = 14$	

9b	$f'(x) = 4x^3 - 12x^2 + 28x - 20$ let $z = f'(x)$ $\frac{dz}{dx} = 12x^2 - 24x + 28$ $= 12\left(x^2 - 2x + \frac{7}{3}\right)$ $= 12(x-1)^2 + 16 > 0$ since $\frac{dz}{dx} > 0$, $z = f'(x)$ is increasing.	- Students need to be very clear what they need to show in the question. In this case, there is a need to show that the derivative of $f'(x)$ [i.e. $f''(x)$] is positive or negative Common errors: - Many substituted $x = 1$ into $f'(x)$ to show $f'(x) = 0$ which is the 1st derivative test and not used to show that a function is increasing / decreasing - Students who correctly arrived at $f''(x)$, did not realise to complete the square to show increasing function.
10a	$7 = \frac{\tan A + 3}{1 - 3 \tan A}$ $22 \tan A = 4$ $\tan A = \frac{2}{11}$ $\cos A = \frac{11}{\sqrt{2^2 + 11^2}}$ $= \frac{11}{5\sqrt{5}}$ $= \frac{11\sqrt{5}}{25}$ Alternative: Cosine Rule  $4^2 = (7^2 + 1^2) + (3^2 + 1^2) - 2(\sqrt{7^2 + 1^2})(\sqrt{3^2 + 1^2}) \cos A$ $16 = 60 - 2\sqrt{500}(\cos A)$ $\cos A = \frac{-44}{-20\sqrt{5}} \left(\frac{\sqrt{5}}{\sqrt{5}} \right)$ $= \frac{11\sqrt{5}}{25}$	
10b	$\sin 2B = 2 \sin B \cos B$ $= 2 \left(\frac{3}{\sqrt{10}} \right) \left(\frac{1}{\sqrt{10}} \right)$ $= \frac{3}{5}$	

11a	$\frac{r}{9} = \frac{30-h}{30}$ $h = 30 - \frac{10}{3}r$ $V = \pi r^2 h$ $= \pi r^2 \left(30 - \frac{10}{3}r \right)$ $= 30\pi r^2 - \frac{10}{3}\pi r^3$	Common errors: $\frac{r}{9} = \frac{h}{30}$ $\frac{r}{18} = \frac{30-h}{30}$ (incorrect use of diameter for ratio) Presentation: Ratios of similar figures/solids should be written in corresponding manner i.e. not $\frac{r}{30-h} = \frac{9}{30}$
11b	$\frac{dV}{dr} = 60\pi r - 10\pi r^2$ Let $\frac{dV}{dr} = 0$, $10\pi r(6-r) = 0$ $r = 0 \text{ (rej.) or } r = 6$ $V = 360\pi$ $= 1130 \text{ cm}^3 \text{ (3 s.f.)}$	Presentation: Include units and label working clearly.
12a	$m_{AC} = \frac{2-7}{0-10}$ $= -\frac{1}{2}$ $m_{BD} = 2$ equation of BD : $y = 2x + 2$	- Practice more on questions with special quadrilaterals to familiarise with properties of these shapes. DO NOT use result from next part to find the answer in the previous part. Method is usually simpler and does not require result from next part.
12b	Method 1A: Use equal adjacent sides of kite $BC = CD$ $(y-2)^2 + (x-10)^2 = 10^2$ $y^2 - 4y + x^2 - 20x + 4 = 0$ -----(1) $y = 2x + 2$ -----(2) sub. (2) into (1), $5x^2 - 20x = 0$ $x^2 - 4x = 0$ $x(x-4) = 0$ $x = 0$ or $x = 4$ $y = 10$ $\therefore B(4,10)$ Method 1B: Use equal adjacent sides of kite Using (a), let $B = (k, 2k+2)$ $AB = AD$ $(2k+2-7)^2 + (k-0)^2 = AD^2$ $(2k-5)^2 + k^2 = 5^2$ $4k^2 - 20k + 25 + k^2 - 25 = 0$ $5k(k-4) = 0$ $k = 0$ (rej.) or $k = 4$ $\therefore B(4,10)$	Many methods to make use of (a) are valid because of the symmetry of the kite e.g. using shoelace method to find area, gradient of perpendicular lines, equation of AC to find midpoint then deduce B, distance of AB o.e.

Method 2: Use shoelace formula & area of triangleADC = area of triangle ABCUsing (a), let $B = (k, 2k + 2)$

$$\frac{1}{2} \begin{vmatrix} 0 & 10 & k & 0 & 0 \\ 2 & 2 & 2k+2 & 7 & 2 \end{vmatrix} = 2 \times \text{Area of } \triangle ADC$$

$$\frac{1}{2} [10(2k+2) + 7k - 2k - 20] = 2 \times \frac{1}{2} \times 5 \times 10$$

$$25k = 100$$

$$k = 4$$

$$\therefore B(4, 10)$$

Method 3: Intersection of AC and BD is midpt. of BD

$$y = -0.5x + 7 \text{ -----(1)}$$

$$y = 2x + 2 \text{ -----(2)}$$

$$(2) = (1),$$

$$-0.5x + 7 = 2x + 2$$

$$2.5x = 5$$

$$x = 2$$

$$y = 6$$

$$\text{Let } B = (x_B, y_B)$$

$$\left(\frac{0 + x_B}{2}, \frac{2 + y_B}{2} \right) = (2, 6)$$

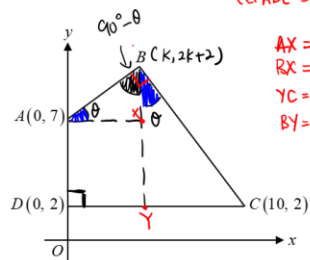
$$x_B = 4$$

$$y_B = 10$$

$$\therefore B(4, 10)$$

Method 4A: Using trigonometry

Since $\angle ADC = 90^\circ$, $\angle ABC = 90^\circ$
 $(\triangle ADC \cong \triangle ABC)$



$$\begin{aligned} AX &= k \\ RX &= 2k - 5 \\ YC &= 10 - k \\ BY &= 2k \end{aligned}$$

$$\tan \angle BAX = \tan \angle CBY$$

$$\frac{2k-5}{k} = \frac{10-k}{2k}$$

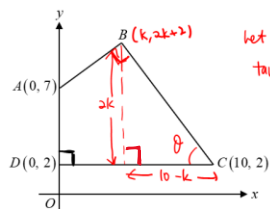
$$4k^2 - 10k = 10k - k^2$$

$$5k^2 - 20k = 0$$

$$5k(k-4) = 0$$

$$k = 0 \text{ (reject)} \text{ or } k = 4$$

$$\therefore B(4, 10)$$

Method 4B: Using trigonometry

$$\text{Let } \angle ACD = \theta.$$

$$\tan \theta = \frac{AD}{DC} = \frac{1}{2}.$$

$$\theta = \tan^{-1}\left(\frac{1}{2}\right).$$

$$\angle BCD = 2\theta$$

$$\therefore \tan 2\theta = \frac{2k}{10-k}$$

$$\frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{2k}{10-k}$$

$$\frac{2(\frac{1}{2})}{1 - (\frac{1}{2})^2} = \frac{2k}{10-k}$$

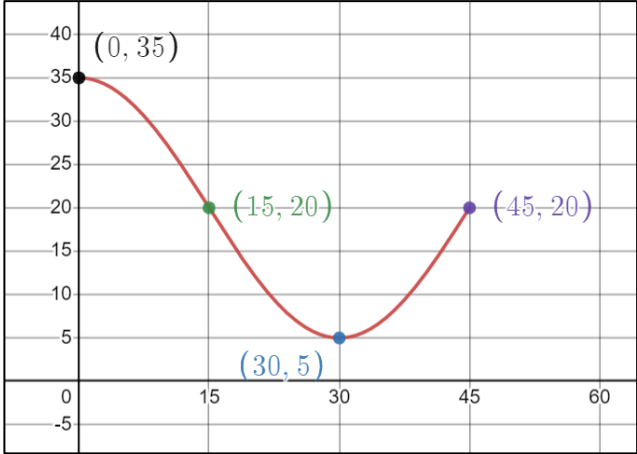
$$10 - k = \frac{3}{4}(2k)$$

$$40 - 4k = 6k.$$

$$10k = 40$$

$$k = 4$$

$$\therefore B(4, 10)$$

12c	Equation of perpendicular bisector of BD is $y = -0.5x + 7$ Equation of perpendicular bisector of AD is $y = 4.5$ Solving simultaneously, $x = 5$ Hence coordinates of point equidistant from A, B and D is $(5, 4.5)$ Alternative method Since angle ABC and angle ADC are 90°, $ABCD$ is a cyclic quadrilateral such that AC is the diameter. Hence the point equidistant from A, B and D is the midpoint of AC, which is $(5, 4.5)$	
13a	$a = \frac{35 - 5}{2}$ $= 15$ $b = \frac{35 + 5}{2}$ $= 20$	Useful relations in trigo graphs for $h = a \cos kt + b$ $a = \frac{\text{max.} - \text{min.}}{2}$ $b = \frac{\text{max.} + \text{min.}}{2}$ $\text{Period} = \frac{2\pi}{k}$
13b	$\text{period} = \frac{2\pi}{k}$ $60 = \frac{2\pi}{k}$ $k = \frac{2\pi}{60}$ $= \frac{\pi}{30} \text{ (shown)}$	
13c		Pay attention to range of values of t. As period = 60 min, if $0 \leq t \leq 45$, hence graph to be sketched is $\frac{3}{4}$ cycle
13d	$10 = 15 \cos \frac{\pi}{30} t + 20$ $\cos \frac{\pi}{30} t = -\frac{2}{3}$ Basic angle = 0.84106 $0 \leq t \leq 60 \Rightarrow 0 \leq \frac{\pi}{30} t \leq 2\pi$ $\frac{\pi}{30} t = \pi - 0.84106, \pi + 0.84106$ $= 2.3005, 3.9826$ $t = 21.968, 38.031$ length of time = $38.031 - 21.968$ = 16.1 minutes (3 s.f.)	Remember to find angles in all relevant quadrants when solving trigo equations to avoid missing out answers, then rejecting answers that are out of range or irrelevant.